



EXPLAINABLE AI DRIVEN BONE TUMOR DETECTION USING RESNET, EFFICIENT NET AND VIT MODELS



PROJECT REPORT

Submitted by

MADHUMITHA R (622522104028)

SHOBANA R (622522104047)

SHRIMATHI K (622522104048)

VIDHYA G (622522104059)

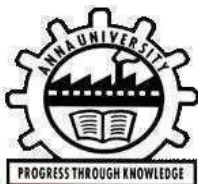
in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



SELVAM COLLEGE OF TECHNOLOGY

(AUTONOMOUS INSTITUTION)

NAMAKKAL

ANNA UNIVERSITY: CHENNAI - 600025

MARCH - 2026



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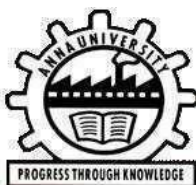
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BONAFIDE CERTIFICATE

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Certified that this project report “**EXPLAINABLE AI-DRIVEN BONE TUMOR USING DETECTION USING HYBRID RESNET, EFFICIENT NET AND VIT MODELS**” is the Bonafide work of **MADHUMITHA R(622522104048), SHOBANA R(622522104047), SHRIMATHI K (622522104048), VIDHYA G (622522104059)** who carried out the Project work under my supervision certified further, that to the best of my knowledge the work reported here in does not from part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Mrs.R.Tamilselvi, M.E.,

Head of the Department,

Department of Computer Science and Engineering,

Selvam College of Technology,

Salem Road (NH-47),

Namakkal -637 003.

SIGNATURE

Mrs.R.Bhuvaneswari, M.E.,

Assistant Professor,

Department of Computer Science and Engineering,

Selvam College of Technology,

Salem Road (NH-47),

Namakkal - 637 003.

Submitted to Project Viva-Voice Examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

ACKNOWLEDGEMENT

With pleasure, I express my sincere gratitude to our Honorable Chairman **Dr.S.BABU, M.B.B.S.**, beloved Secretary **Dr.B.KAVIETHRANANDHINI, B.E.,M.S.,M.B.A.,Ph.D.**,and our beloved Executive Director **Mr.M.KARTHIK, MCA., CCNA.**, for giving motivation and providing all necessary facility for the successful completion of the project.

It is my privilege to thank our Respected Principal **Dr.A.JEGAN, M.E., Ph.D.**, for the moral support and deeds in bringing out the project successfully.

I would like to thank our beloved Head, Department of Computer Science and Engineering **Mrs.R.TAMILSELVI, M.E.**, for providing me with this wonderful opportunity to do this project and encouraged me to complete it successfully.

I express my sincere thanks to my Project Coordinator **Ms.S.SARANYA, M.E.**, for his continuous support and timely help during the project work.

I also thank my project guide **Mrs.R.BHUVANESWARI,M.E.**, Assistant professor for her helpful hands in each step of my project and for this guidance and constant support in the project work.

I express my thanks to all my department faculty members, friends, and family members for their advice and encouragement to do the project with full interest and enthusiasm.

ABSTRACT

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Bone tumors are abnormal growths of cells within bone tissue that can be either benign or malignant. Early and accurate detection of bone tumors is essential for effective treatment and improved patient outcomes. However, manual analysis of medical images such as X-rays or MRI scans can be time-consuming and may sometimes lead to diagnostic errors due to human limitations. This project proposes an Explainable AI-Driven Bone Tumor Detection System that utilizes hybrid deep learning models to improve diagnostic accuracy. The system integrates advanced architectures such as ResNet-50, EfficientNet, and Vision Transformer (ViT) to effectively extract spatial and contextual features from bone images. Prior to model training, medical images undergo preprocessing techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) and morphological operations to enhance image quality and highlight tumor regions. Transfer learning is applied to leverage pretrained models, reducing computational complexity while improving detection performance. An ensemble learning strategy combines the predictions from multiple deep learning models to increase classification reliability. The system provides a user-friendly interface where users can upload bone images and receive prediction results along with visual explanations. The proposed approach aims to assist medical professionals in early detection and diagnosis of bone tumors. By combining deep learning accuracy with explainable AI transparency, the system enhances trust and usability in clinical environments. Overall, this research demonstrates the potential of artificial intelligence in improving medical imaging analysis and supporting intelligent healthcare systems.

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Bone tumors are abnormal growths that occur in bone tissues and can be classified as either benign or malignant. Early detection of bone tumors is essential for effective treatment planning and improving patient survival rates. Medical imaging techniques such as X-ray, Computed Tomography (CT), and Magnetic Resonance Imaging (MRI) are widely used by radiologists to examine bone abnormalities and detect tumor presence. However, manual interpretation of these medical images is often time-consuming, requires significant expertise, and may lead to inconsistencies due to human subjectivity. In many healthcare environments, especially in developing regions, the shortage of experienced radiologists further complicates timely diagnosis.

In recent years, Artificial Intelligence (AI) and Deep Learning have significantly transformed the field of medical image analysis. Deep learning models, particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable capability in automatically extracting complex features from medical images and performing accurate classification tasks. These models have been successfully applied in various diagnostic systems such as tumor detection, disease classification, and medical image segmentation. Despite these advancements, many existing AI-based systems for bone tumor detection still face limitations such as insufficient accuracy, limited generalization across datasets, and lack of interpretability.

Traditional CNN-based models primarily focus on extracting local spatial features from images, which may not fully capture the global contextual relationships present in medical imaging data. As a result, subtle tumor patterns or irregular bone structures may not be effectively identified. Additionally, many deep learning models function as “black boxes,” meaning their internal decision-making processes are not easily understandable to clinicians. This lack of transparency creates hesitation among healthcare professionals to rely fully on AI-based diagnostic systems.

To overcome these limitations, this project proposes an Explainable AI-driven hybrid deep learning framework for bone tumor detection using ResNet-50, EfficientNet, and Vision Transformer (ViT) models. The proposed system integrates multiple advanced deep learning architectures to improve feature extraction and classification performance. ResNet-50 and EfficientNet are utilized for efficient convolutional feature extraction, while Vision Transformers capture long-range dependencies and global contextual information through attention mechanisms. By combining these models within a hybrid framework, the system can analyze both local and global patterns present in medical images.

Another key component of the proposed framework is the use of advanced image preprocessing techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) and adaptive morphological operations. These techniques enhance the quality of medical images by improving contrast, reducing noise, and highlighting potential tumor regions. Improved image quality allows the deep learning models to learn more discriminative features and improve diagnostic accuracy.

In addition to improving performance, the proposed system emphasizes explainability and transparency, which are crucial in medical AI applications. Grad-CAM (Gradient-weighted Class Activation Mapping) visualization techniques are

integrated into the framework to generate heatmaps that highlight regions of the image influencing the model's prediction. This explainable AI approach allows radiologists to understand how the model identifies tumor regions, thereby increasing trust and reliability in AI-assisted diagnostic systems.

Overall, the proposed hybrid deep learning framework aims to provide a robust, interpretable, and efficient solution for automated bone tumor detection. The system can serve as a decision-support tool for medical professionals, assisting them in early diagnosis, reducing diagnostic workload, and improving patient outcomes.

1.1.1 Background of the Study

Bone tumors represent a significant medical concern due to their potential to disrupt bone structure, cause pain, and in severe cases, spread to other parts of the body. Early diagnosis plays a critical role in determining the appropriate treatment strategy, whether surgical removal, chemotherapy, or radiation therapy. Traditionally, radiologists analyze medical images manually to identify abnormal bone growth patterns, irregular textures, or suspicious lesions. While this method has been effective for many years, it is highly dependent on human expertise and can sometimes lead to diagnostic errors.

The development of Computer-Aided Diagnosis (CAD) systems has helped support radiologists by highlighting suspicious areas in medical images. Early CAD systems relied on handcrafted features extracted from images, including shape descriptors, texture patterns, and intensity distributions. These features were then used in machine learning algorithms such as Support Vector Machines (SVMs) or decision trees to classify images. However, handcrafted feature extraction methods often lack flexibility and may fail to capture complex structures present in medical images.

The emergence of deep learning has revolutionized medical imaging by enabling automatic feature extraction directly from raw images. Convolutional Neural Networks (CNNs) have been widely used in medical image classification tasks because of their ability to learn hierarchical feature representations. Popular CNN architectures such as VGGNet, ResNet, and DenseNet have shown promising performance in various disease detection applications.

Despite their success, CNN-based models have certain limitations. They primarily capture local spatial features through convolutional operations and may struggle to model long-range relationships within images. Medical images often contain complex structures where contextual information from distant regions can be crucial for accurate diagnosis. As a result, relying solely on CNN architectures may limit the system's ability to capture comprehensive image features.

To address these challenges, Vision Transformers (ViTs) have recently been introduced in computer vision research. Unlike CNNs, Vision Transformers utilize self-attention mechanisms that allow the model to capture global contextual relationships between different image regions. This capability makes them particularly useful in medical imaging tasks where understanding spatial relationships is important.

Another major challenge in medical image analysis is model interpretability. Many deep learning systems produce predictions without providing explanations, making them difficult for clinicians to trust. Explainable AI techniques such as Grad-CAM provide visual explanations by highlighting important image regions responsible for predictions. Integrating explainability into diagnostic systems improves transparency and supports clinical validation.

1.1.2 Need for the Proposed System

Although significant progress has been made in AI-based medical imaging systems, several limitations still exist in current bone tumor detection frameworks. Many existing systems rely on single CNN architectures that focus primarily on local spatial patterns and may fail to capture complex contextual relationships within medical images. Additionally, basic preprocessing techniques used in traditional systems may not adequately enhance tumor regions, leading to reduced detection accuracy.

Another major challenge is the lack of interpretability in deep learning models. Most AI systems operate as black-box models, providing predictions without clear explanations. This lack of transparency reduces the level of trust among medical professionals and limits the practical adoption of AI-based diagnostic tools in clinical environments.

Furthermore, many existing models struggle with generalization when applied to different datasets or imaging conditions. Variations in imaging devices, patient anatomy, and scanning techniques can affect model performance. Therefore, there is a need for a more robust and adaptable framework that can maintain high accuracy across diverse medical imaging scenarios.

The proposed system addresses these challenges by integrating hybrid deep learning architectures that combine the strengths of CNN-based models and Vision Transformers. Advanced preprocessing techniques improve image quality, while explainable AI methods provide transparency in decision-making. This integrated approach aims to improve diagnostic accuracy, enhance interpretability, and increase clinical acceptance of AI-assisted medical diagnosis.

1.1.3 Objectives within the Overview Context

The primary objective of this project is to develop an advanced and interpretable AI-based system for automated bone tumor detection using hybrid deep learning models. The specific objectives include:

- Improving bone tumor detection accuracy using hybrid deep learning architectures
- Integrating ResNet-50, Efficient Net, and Vision Transformer models for enhanced feature extraction
- Enhancing medical image quality using advanced preprocessing techniques such as CLAHE
- Implementing explainable AI techniques such as Grad-CAM for transparent model predictions
- Developing a reliable decision-support tool for radiologists and healthcare professionals

1.1.4 Significance of the Project

The proposed explainable AI-driven bone tumor detection system contributes significantly to the advancement of intelligent healthcare technologies. By combining powerful deep learning models with explainable AI techniques, the system not only achieves high diagnostic performance but also ensures transparency and interpretability in predictions. This is particularly important in medical applications where trust and reliability are essential.

The hybrid integration of ResNet, EfficientNet, and Vision Transformers enables the system to analyze both local and global features within medical images, leading to improved classification accuracy. Additionally, the use of advanced

preprocessing techniques enhances the visibility of tumor regions, allowing the model to detect subtle abnormalities more effectively.

From a practical perspective, the system can act as a **computer-aided diagnostic tool** that assists radiologists in identifying bone tumors more efficiently. It can help reduce diagnostic workload, minimize human errors, and support early disease detection. Furthermore, the explainability component allows clinicians to validate AI predictions, thereby increasing confidence in the system.

Ultimately, this project contributes to the development of reliable AI-assisted medical diagnostic systems that improve healthcare delivery and patient outcomes. By supporting early detection and accurate diagnosis, the system aligns with global healthcare goals aimed at improving patient care and reducing mortality associated with cancer-related diseases.

1.2 PROBLEM IDENTIFICATION

The scope of the proposed project focuses on developing an intelligent and explainable deep learning–based system for automated bone tumor detection using medical images. The system aims to assist medical professionals in identifying potential bone tumors by analyzing X-ray or MRI images through advanced artificial intelligence techniques. With the increasing demand for accurate and fast diagnostic tools in healthcare, this project emphasizes the application of hybrid deep learning models and explainable AI methods to improve the reliability and interpretability of medical image analysis.

The project primarily deals with the processing and analysis of bone medical images using modern deep learning architectures such as ResNet-50, EfficientNet, and Vision Transformer (ViT). These models are integrated within a hybrid

framework to extract both local spatial features and global contextual information from medical images. The use of multiple deep learning architectures allows the system to learn complex patterns associated with tumor regions and improves the overall classification performance. By combining convolution-based feature extraction with transformer-based attention mechanisms, the proposed framework aims to provide a more comprehensive understanding of image characteristics compared to traditional CNN-only systems.

Another important aspect within the scope of the project is image preprocessing and enhancement. Medical images often suffer from noise, low contrast, and variations caused by different imaging devices or environmental conditions. To address these challenges, advanced preprocessing techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) and adaptive morphological operations are incorporated into the system. These techniques improve image quality by enhancing contrast and highlighting important structural details in bone images, which ultimately helps the deep learning models detect tumor regions more effectively.

The project also focuses on implementing transfer learning techniques to improve model efficiency and reduce training complexity. Transfer learning enables the system to utilize pretrained models that have already learned general image features from large datasets. By fine-tuning these models on bone tumor datasets, the system can achieve high accuracy even with limited medical imaging data. This approach significantly reduces computational requirements and improves model convergence during training.

Explainable Artificial Intelligence (XAI) forms another critical component within the scope of the project. In medical applications, it is essential for AI systems to provide transparent and interpretable predictions. Therefore, the proposed system

integrates visualization techniques such as Grad-CAM to generate heatmaps that highlight the regions of medical images responsible for the model's predictions. These visual explanations allow radiologists and medical professionals to understand the reasoning behind the system's decisions, thereby increasing trust and acceptance of AI-assisted diagnostic tools.

Furthermore, the project includes the development of a user-friendly interface that allows users to upload medical images and receive tumor detection results along with visual explanations. The results generated by the system are intended to assist healthcare professionals in identifying suspicious bone regions more efficiently. The system acts as a supportive diagnostic tool rather than replacing medical experts, ensuring that final clinical decisions remain under the supervision of qualified healthcare professionals.

The scope of the project also includes evaluating the performance of the proposed hybrid deep learning framework using standard evaluation metrics such as accuracy, precision, recall, and F1-score. These metrics help measure the effectiveness of the system in detecting bone tumors and allow comparison with traditional CNN-based models. Through experimental evaluation, the project aims to demonstrate that the hybrid architecture improves classification accuracy and provides better generalization across medical imaging datasets.

However, the scope of the project is limited to the detection and classification of bone tumors from medical images. It does not include clinical treatment planning, tumor staging, or prediction of patient survival outcomes. Additionally, the system relies on available datasets for model training and evaluation, which may influence performance depending on dataset quality and diversity.

Overall, the project focuses on designing and implementing a reliable, interpretable, and efficient AI-based framework for bone tumor detection. By integrating hybrid deep learning models, advanced image preprocessing, and explainable AI techniques, the system aims to support radiologists in early diagnosis and contribute to improved healthcare outcomes. The proposed framework also lays the foundation for future research in intelligent medical image analysis and AI-assisted clinical decision support systems.

1.3 DEEP LEARNING

Deep Learning is a specialized branch of Artificial Intelligence (AI) and Machine Learning (ML) that focuses on building multi-layered neural network models capable of learning complex patterns from large amounts of data. Unlike traditional machine learning algorithms that depend on manually designed features, deep learning models automatically learn hierarchical feature representations directly from raw data. These models use multiple hidden layers to progressively extract low-level, mid-level, and high-level features, enabling them to perform complex tasks such as image classification, object detection, speech recognition, and natural language understanding.

In recent years, deep learning has become a dominant technology in the field of computer vision and medical image analysis. Medical imaging modalities such as X-ray, MRI, CT scan, and ultrasound generate large volumes of visual data that require careful interpretation by experienced radiologists. Deep learning techniques help automate this process by learning patterns associated with diseases directly from medical images. By analyzing pixel-level information and spatial patterns, deep learning models can detect abnormalities, classify diseases, and assist healthcare professionals in clinical decision-making.

The effectiveness of deep learning in medical imaging lies in its ability to learn both **low-level and high-level image features**. Low-level features include edges, textures, and intensity variations, while high-level features capture complex patterns such as shapes, structures, and anatomical relationships. This hierarchical learning capability allows deep neural networks to identify subtle variations in medical images that may not be easily noticeable to the human eye.

Deep learning models have been widely applied in healthcare applications such as tumor detection, disease classification, organ segmentation, and medical report generation. In bone tumor detection systems, deep learning algorithms analyze bone X-ray or MRI images to identify irregular bone structures, abnormal growth patterns, and suspicious lesions. These models significantly reduce diagnostic workload and improve the speed and accuracy of disease detection.

Another important advantage of deep learning is its ability to improve performance with larger datasets. As the amount of training data increases, deep neural networks become better at generalizing patterns and making accurate predictions. Furthermore, the use of pretrained models and transfer learning allows researchers to leverage existing knowledge learned from large datasets, thereby improving performance even when medical datasets are limited.

Modern deep learning research also focuses on improving model interpretability and transparency. In healthcare applications, it is important for AI models to provide explanations for their predictions. Explainable AI techniques such as Grad-CAM allow researchers and clinicians to visualize the regions of medical images that influence the model's decisions. These techniques help increase trust in AI-assisted diagnostic systems and support their adoption in clinical environments.

Overall, deep learning has become an essential technology in medical image analysis due to its ability to learn complex representations, process large datasets, and provide accurate predictions. By combining different deep learning architectures, such as Convolutional Neural Networks and Transformer-based models, researchers can develop powerful hybrid systems that improve disease detection and diagnostic accuracy.

1.3.1 Artificial Neural Networks (ANN)

Artificial Neural Networks (ANNs) are computational models inspired by the structure and functioning of the human brain. They consist of interconnected processing units called neurons that work together to process information and solve complex problems. Each neuron receives input signals, performs a mathematical operation, and produces an output that is passed to other neurons in the network.

An ANN typically consists of three main layers: the **input layer**, **hidden layers**, and **output layer**. The input layer receives the raw data, such as pixel values from images. The hidden layers perform complex computations and extract features from the input data. The output layer produces the final prediction or classification result.

Each neuron in the network applies weights and biases to the input signals and processes them through an activation function. Activation functions introduce non-linearity into the network, enabling the model to learn complex relationships between inputs and outputs. Common activation functions used in neural networks include ReLU (Rectified Linear Unit), sigmoid, and tanh.

The learning process in neural networks occurs through a training mechanism known as **backpropagation**. During training, the network makes predictions and compares them with actual target values. The difference between predicted and actual

outputs is measured using a loss function. The backpropagation algorithm then adjusts the weights of the network to minimize this error, improving prediction accuracy over time.

Although Artificial Neural Networks are powerful tools for pattern recognition and classification, shallow neural networks with only one or two hidden layers have limited capability when dealing with high-dimensional data such as images. This limitation led to the development of deep neural networks with multiple hidden layers, enabling more sophisticated feature extraction and hierarchical learning.

1.3.2 Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNNs) are specialized deep learning architectures designed for image processing and computer vision tasks. Unlike traditional neural networks, CNNs are capable of automatically learning spatial hierarchies of features from images using convolution operations.

The fundamental idea behind CNNs is to use small learnable filters, known as kernels, to scan across the input image. These filters detect patterns such as edges, textures, and shapes by performing convolution operations. The resulting feature maps capture important visual information that is used for further processing within the network.

A typical CNN architecture consists of several key components:

Convolutional Layer: This layer performs convolution operations using filters to extract feature maps from the input image. Each filter learns to detect specific visual patterns.

Pooling Layer: Pooling layers reduce the spatial dimensions of feature maps, which helps decrease computational complexity and prevent overfitting. Common pooling operations include max pooling and average pooling.

Activation Function: Activation functions introduce non-linearity into the network, allowing the model to learn complex patterns. ReLU is the most widely used activation function in CNN architectures.

Fully Connected Layer: This layer connects all neurons from the previous layer and performs the final classification task by mapping extracted features to output classes.

CNNs have achieved remarkable success in medical image classification because they can automatically extract meaningful features without manual intervention. Popular CNN architectures such as **VGGNet, ResNet, DenseNet, and EfficientNet** have been widely used in medical imaging applications.

However, CNNs mainly focus on **local spatial features** due to their limited receptive fields. While they are effective in detecting small patterns, they may struggle to capture global contextual relationships within images. This limitation has motivated researchers to explore new architectures that can capture long-range dependencies in image data.

1.3.3 Vision Transformers (ViT)

Vision Transformers represent a recent breakthrough in deep learning for image analysis. Inspired by the success of transformer models in natural language processing, Vision Transformers apply the self-attention mechanism to visual data,

enabling the model to capture global contextual relationships across different parts of an image.

Unlike CNNs that process images using convolutional filters, Vision Transformers divide the input image into smaller patches. Each patch is then converted into a vector representation known as a patch embedding. These embeddings are passed through multiple transformer encoder layers, which apply self-attention operations to learn relationships between different image regions.

The self-attention mechanism allows the model to assign importance weights to different patches of the image. By analyzing the interactions between patches, Vision Transformers can capture long-range dependencies and understand the global structure of images more effectively.

One of the main advantages of Vision Transformers is their ability to model **global context**. In medical imaging applications, disease patterns may span across multiple regions of an image, making global contextual information essential for accurate diagnosis. Vision Transformers address this challenge by learning relationships between distant regions within the image.

Although Vision Transformers provide strong contextual understanding, they typically require large datasets for training. Therefore, combining CNN-based feature extractors with Vision Transformers creates a hybrid framework that leverages the strengths of both architectures. CNNs efficiently capture local spatial patterns, while Vision Transformers provide global contextual awareness.

This hybrid approach is particularly beneficial in medical imaging applications such as bone tumor detection, where both local texture patterns and global structural information are important for accurate classification.

1.3.4 Explainable AI in Deep Learning

Explainable Artificial Intelligence (XAI) refers to techniques that make the decisions of deep learning models transparent and interpretable. In many deep learning applications, models function as black boxes, meaning they provide predictions without explaining the reasoning behind them. While this may be acceptable in some domains, it poses serious challenges in healthcare, where clinicians require clear explanations before trusting AI-based diagnostic systems.

Explainable AI techniques aim to bridge this gap by providing visual or mathematical explanations for model predictions. One widely used method in medical imaging is **Gradient-weighted Class Activation Mapping (Grad-CAM)**. Grad-CAM generates heatmaps that highlight the regions of an image that contributed most to the model's prediction.

In bone tumor detection systems, Grad-CAM can visually indicate the specific bone regions where the model detects abnormal patterns or tumor structures. These visual explanations help radiologists verify whether the AI system is focusing on medically relevant areas.

The integration of explainable AI techniques enhances transparency, increases clinician trust, and ensures that AI systems can be safely used in real-world medical environments. By combining powerful deep learning models with explainability mechanisms, researchers can develop intelligent diagnostic systems that are both accurate and interpretable.

1.4 AIM AND OBJECTIVES

1.4.1 AIM

The aim of this project is to develop an Explainable Artificial Intelligence (XAI) based bone tumor detection system using hybrid deep learning models such as ResNet, EfficientNet, and Vision Transformer (ViT). The system aims to analyze medical images such as X-rays or MRI scans to automatically detect and classify bone tumors with high accuracy. Additionally, the system integrates explainability techniques like Grad-CAM to highlight tumor regions and provide interpretable results, assisting medical professionals in making reliable and faster diagnostic decisions.

1.4.2 OBJECTIVES

- To develop an automated system for detecting bone tumors from medical images using deep learning techniques.
- To preprocess medical images using techniques such as CLAHE and morphological operations to enhance image quality and highlight tumor regions.
- To implement ResNet-50 and EfficientNet architectures for effective feature extraction from bone images.
- To integrate Vision Transformer (ViT) for capturing global contextual relationships within medical images.
- To design a hybrid deep learning framework that combines CNN-based models and transformer models for improved classification performance.
- To incorporate Explainable AI techniques such as Grad-CAM to visualize the regions influencing the model's predictions.

- To evaluate the performance of the proposed system using metrics such as accuracy, precision, recall, and F1-score.

1.5 SCOPE OF THE PROJECT

The scope of this project focuses on the development of an intelligent and explainable deep learning–based system for the automatic detection of bone tumors from medical images. The system aims to assist medical professionals by analyzing X-ray or MRI images and identifying abnormal bone regions that may indicate the presence of tumors. With the growing demand for accurate and efficient diagnostic tools in healthcare, this project utilizes advanced Artificial Intelligence techniques to support early diagnosis and improve clinical decision-making.

The proposed system primarily uses hybrid deep learning architectures such as **ResNet, EfficientNet, and Vision Transformer (ViT)** to extract meaningful features from bone images. These models are designed to capture both local and global image patterns, allowing the system to detect subtle abnormalities in bone structures. The combination of convolutional neural networks and transformer-based models enhances the overall performance of the system by improving feature learning and classification accuracy.

Another important component within the scope of this project is **medical image preprocessing and enhancement**. Techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) and morphological operations are used to improve the quality of input images. These methods enhance image contrast, reduce noise, and highlight tumor regions, which helps the deep learning models learn more discriminative features during training.

The project also includes the implementation of **transfer learning**, which allows the use of pretrained deep learning models to improve performance and reduce training time. By fine-tuning these models using bone tumor datasets, the system can achieve high detection accuracy even with limited medical imaging data.

In addition to accurate detection, the project emphasizes **explainable AI (XAI)**. Techniques such as Grad-CAM are integrated to generate visual heatmaps that highlight the regions of the image influencing the model's predictions. These explanations help doctors understand how the AI system identifies tumor regions, thereby increasing transparency and trust in AI-assisted diagnosis.

The scope of the project also includes the evaluation of the proposed system using performance metrics such as accuracy, precision, recall, and F1-score. However, the project is limited to the detection and classification of bone tumors from medical images and does not cover clinical treatment planning or patient management. Overall, the system aims to serve as a supportive diagnostic tool for healthcare professionals, improving early detection and contributing to better healthcare outcomes.

CHAPTER 2

LITERATURE SURVEY

2.1 Title: Deep Learning-Based Bone Tumor Detection Using Convolutional Neural Networks

Author(s): S. S. Kumar, R. S. Babu

Methodology:

This study presents a deep learning approach for detecting bone tumors using Convolutional Neural Networks (CNN). The researchers focused on improving the accuracy of tumor detection from medical X-ray images by utilizing automatic feature extraction capabilities of CNN architectures. The dataset used in the study consisted of labeled bone X-ray images categorized into normal and tumor classes.

The preprocessing stage involved resizing images, noise removal, and image normalization to ensure consistency across the dataset. The CNN architecture consisted of multiple convolutional layers, pooling layers, and fully connected layers designed to capture spatial features from bone images. Feature extraction was performed automatically through convolution operations, allowing the model to learn texture patterns and abnormal bone structures.

The trained model was evaluated using metrics such as accuracy, precision, recall, and F1-score. The results were compared with traditional machine learning methods that relied on handcrafted features.

Findings:

The CNN-based model significantly improved tumor detection accuracy compared to traditional methods. The study concluded that deep learning provides effective feature extraction and classification capabilities for bone tumor detection, although the model lacked interpretability and explainability mechanisms.

2.2 Title: Medical Image Classification Using EfficientNet Deep Learning Architecture

Author(s): Mingxing Tan, Quoc V. Le

Methodology:

This research introduced EfficientNet, a deep learning architecture designed to improve model performance by optimizing network scaling. EfficientNet uses a compound scaling method that balances network depth, width, and resolution to achieve higher accuracy with fewer parameters.

The study evaluated EfficientNet on multiple medical image classification tasks, including disease detection from medical scans. The dataset included large collections of labeled medical images that were preprocessed using normalization, image resizing, and data augmentation techniques. Transfer learning was applied using pretrained EfficientNet models to improve feature learning efficiency.

The architecture consisted of convolutional layers combined with mobile inverted bottleneck convolution (MBConv) blocks that efficiently extracted hierarchical features from images. The model was trained and validated using standard evaluation metrics such as classification accuracy and loss curves.

Findings:

The EfficientNet model demonstrated superior performance compared to conventional CNN architectures while maintaining lower computational complexity. The study concluded that EfficientNet is highly effective for medical image classification due to its efficient feature extraction capabilities.

2.3 Title: Vision Transformer for Image Recognition at Scale

Author(s): Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov

Methodology:

This research proposed the Vision Transformer (ViT), a novel deep learning architecture designed for image recognition tasks. Unlike traditional CNN models, Vision Transformers process images using a transformer-based self-attention mechanism. The image is divided into smaller patches, and each patch is converted into a vector representation before being passed through transformer encoder layers.

The model was trained on large-scale image datasets to evaluate its performance in image classification tasks. Preprocessing steps included image resizing, patch embedding generation, and positional encoding to preserve spatial information. The transformer encoder layers applied multi-head self-attention to capture relationships between different image patches.

The ViT model was evaluated using accuracy and classification benchmarks against conventional CNN architectures such as ResNet.

Findings:

The Vision Transformer achieved competitive and often superior performance compared to CNN models in large-scale image classification tasks. The study highlighted the ability of transformers to capture global contextual information, which is beneficial in complex image analysis tasks such as medical diagnosis.

2.4 Title: Explainable AI for Medical Imaging Using Grad-CAM Visualization

Author(s): Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das

Methodology:

This research introduced Gradient-weighted Class Activation Mapping (Grad-CAM), an explainable AI technique used to visualize deep learning model decisions. The objective of the study was to improve transparency and interpretability in deep neural networks used for image classification tasks.

Grad-CAM works by computing the gradients of the target class with respect to feature maps in the final convolutional layer of a CNN model. These gradients are then used to generate heatmaps that highlight the most important regions of the image responsible for the prediction.

The method was tested on multiple deep learning architectures including VGGNet, ResNet, and DenseNet. The researchers applied Grad-CAM visualization on medical image datasets to identify disease regions and analyze model behavior.

Findings:

The Grad-CAM technique successfully provided visual explanations for deep learning predictions. The study concluded that explainable AI techniques significantly improve trust and reliability in AI-based medical diagnosis systems.

2.5 Title: Hybrid Deep Learning Framework for Medical Image Classification

Author(s): Y. Li, H. Zhang, and X. Chen

Methodology:

This study proposed a hybrid deep learning framework that combines Convolutional Neural Networks and transformer-based architectures for improved medical image classification. The objective was to leverage the strengths of CNN models in extracting local spatial features and transformers in capturing global contextual relationships.

The dataset used in the study consisted of multiple medical imaging modalities such as X-rays and MRI scans. The preprocessing stage included image normalization, resizing, and contrast enhancement. A CNN-based backbone model such as ResNet was used to extract spatial features from the images. These features were then passed to a transformer module to capture global dependencies across image regions.

An ensemble strategy was implemented to combine predictions from both CNN and transformer components. The system was evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score.

Findings:

The hybrid CNN–Transformer framework achieved improved classification accuracy compared to standalone CNN models. The study demonstrated that combining local feature extraction with global contextual understanding enhances medical image analysis performance.

CHAPTER 3

SYSTEM SPECIFICATION

3.1. HARDWARE REQUIREMENTS

- **Processor** : Intel® Core™ i5 processor or higher (i7 recommended)
- **RAM** : 16 GB (32 GB recommended for deep learning training)
- **Disk Space** : 500 GB SSD (for datasets and trained models)
- **GPU** : NVIDIA GPU with CUDA support (Recommended for model training)
- **Display** : Standard Monitor

3.2. SOFTWARE REQUIREMENTS

1. Operating System:

- Windows 10 / Windows 11
- Ubuntu Linux (Optional for better deep learning support)

2. Application Development:

- Python for backend development
- Flask as the web framework for building the diagnostic web application

3. Database:

- MySQL for storing patient image records, prediction results, and system data
- SQLite (Optional for lightweight deployment)

4. Machine Learning and Deep Learning:

- TensorFlow / PyTorch for deep learning model development
- Scikit-Learn for ensemble learning (XGBoost integration)
- OpenCV for image preprocessing and enhancement
- NumPy for numerical computations

5. Data Analysis and Visualization:

- Pandas for dataset handling and preprocessing
- Matplotlib and Seaborn for performance visualization
- Scikit-Plot for model evaluation metrics

6. Development Environment:

- Jupyter Notebook for experimentation
- VS Code / PyCharm for development
- Anaconda for environment management

7. Visualization and Explainability:

- Grad-CAM++ implementation libraries
- Transformer Attention Visualization tools

CHAPTER 4

SYSTEM ANALYSIS

4.1 EXISTING SYSTEM

The detection and diagnosis of bone tumors in the existing healthcare system are primarily carried out using conventional medical imaging techniques and manual analysis performed by radiologists and orthopedic specialists. Common imaging methods used include X-ray imaging, Magnetic Resonance Imaging (MRI), Computed Tomography (CT) scans, and biopsy tests. These techniques help doctors identify abnormal growths in bones and determine whether the tumor is benign or malignant.

In the current system, radiologists examine medical images manually to identify suspicious regions that may indicate the presence of bone tumors. The diagnosis process requires significant expertise and experience because bone tumors can appear in different shapes, sizes, and locations. Radiologists must carefully analyze various features in the image such as bone density, texture, shape irregularities, and lesion patterns. This process is often time-consuming and may vary depending on the experience level of the medical professional.

Another important step in the existing diagnostic process is biopsy testing. When a suspicious tumor is detected in imaging scans, a small sample of bone tissue is collected from the affected area and analyzed in a laboratory. This procedure helps confirm whether the tumor is cancerous or non-cancerous. However, biopsy procedures are invasive, may cause discomfort to patients, and require additional time for laboratory analysis and results.

Furthermore, the current system relies heavily on the manual interpretation of medical images. Since radiologists must review a large number of scans every day, there is a possibility of human error, especially in early-stage tumor detection where abnormalities may be very small or difficult to identify. Subtle tumor patterns can sometimes be overlooked, which may delay diagnosis and treatment.

The existing system also faces challenges in rural or underdeveloped healthcare centers where experienced radiologists and advanced diagnostic facilities may not be available. In such cases, patients must travel to larger hospitals or specialized diagnostic centers to obtain proper imaging and expert consultation. This can increase both the time and cost involved in diagnosis.

Another limitation is the lack of automated decision-support tools in many healthcare systems. Traditional diagnosis methods do not fully utilize advanced technologies such as Artificial Intelligence (AI), Machine Learning (ML), or deep learning-based image analysis. Without such technologies, it becomes difficult to process large volumes of medical images quickly and consistently.

In addition, manual analysis may lead to variations in diagnosis between different radiologists. Two experts may interpret the same medical image differently depending on their experience and perspective. This inconsistency can sometimes affect the reliability of the diagnosis and may require additional medical opinions.

Overall, the existing system plays a critical role in diagnosing bone tumors, but it mainly depends on manual analysis, invasive procedures, and the availability of expert radiologists. These limitations highlight the need for more advanced, automated, and intelligent systems that can assist doctors in detecting bone tumors more efficiently and accurately.

4.1.1 DISADVANTAGES

- Diagnosis mainly depends on radiologist experience.
- Manual image analysis is time-consuming.
- Possibility of human errors in interpretation.
- Early-stage tumors may be difficult to detect.
- Requires multiple medical imaging tests.
- Biopsy procedures are invasive and painful.

4.2 PROPOSED SYSTEM

The proposed system introduces an advanced and intelligent framework for the automated detection and classification of bone tumors using explainable Artificial Intelligence (AI) and hybrid deep learning models. The system is designed to assist medical professionals in accurately identifying bone tumors from medical imaging data such as X-ray or MRI scans. By leveraging state-of-the-art deep learning architectures and advanced image processing techniques, the proposed system aims to improve diagnostic accuracy, reduce human errors, and accelerate the tumor detection process.

In this proposed framework, multiple deep learning architectures are integrated to enhance feature extraction and classification performance. The system combines powerful convolutional neural network (CNN) architectures such as **ResNet-50** and **EfficientNet** with transformer-based models such as the **Vision Transformer (ViT)**. These architectures work together to capture both local spatial features and global contextual relationships present in medical images.

ResNet-50 is used for its ability to effectively train deep networks using residual connections, which help avoid the vanishing gradient problem. EfficientNet is incorporated because of its efficient scaling strategy that improves model performance while maintaining computational efficiency. Vision Transformers contribute to the model by capturing long-range dependencies within the image through a self-attention mechanism. By combining these architectures, the proposed system benefits from the strengths of both convolutional and transformer-based models.

Before feeding images into the deep learning models, the system performs advanced preprocessing steps to enhance image quality and highlight important tumor-related features. Techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) are applied to improve image contrast and reveal subtle bone abnormalities that may not be visible in raw images. Additionally, adaptive morphological operations are used to enhance edges, remove noise, and emphasize tumor boundaries.

The proposed system also implements an ensemble learning strategy. Instead of relying on a single model, predictions from multiple models are combined to generate a final classification result. This ensemble approach increases the robustness and reliability of the system by reducing the risk of model bias and improving generalization across different types of bone tumor images.

Another important feature of the proposed system is its explainability component. Explainable AI (XAI) techniques such as Grad-CAM or attention visualization can be incorporated to highlight the regions of the image that contribute most to the model's prediction. This helps doctors understand how the system arrives at its decisions and increases trust in the automated diagnosis process.

The system processes medical images through several stages, including image acquisition, preprocessing, feature extraction, tumor detection, and classification. The final output indicates whether the tumor is benign or malignant and highlights the suspicious regions within the image. This assists medical professionals in making faster and more informed clinical decisions.

Overall, the proposed system aims to provide an efficient, accurate, and reliable solution for bone tumor detection. By combining deep learning, image enhancement techniques, and explainable AI mechanisms, the system supports early diagnosis and improves the effectiveness of medical image analysis.

4.2.1 ADVANTAGES

- Provides automated bone tumor detection.
- Improves diagnostic accuracy using deep learning models.
- Combines CNN and Transformer architectures.
- Captures both local and global image features.
- Reduces human errors in tumor identification.
- Enhances image quality using preprocessing techniques.
- Faster analysis of medical images.

4.3 SYSTEM STUDY

System study is an important phase in the development of any software or research-based system. It involves analyzing the existing methods, identifying limitations, understanding user requirements, and determining how the proposed system can overcome the current challenges. In the context of medical image analysis, system study helps in understanding how bone tumors are currently detected and how artificial intelligence can improve the diagnosis process.

Bone tumors are abnormal growths of cells within bone tissue that can be either benign (non-cancerous) or malignant (cancerous). Early detection of bone tumors is essential for effective treatment and improved patient survival rates. Traditionally, bone tumor detection is performed by radiologists through the manual analysis of medical images such as X-rays, CT scans, or MRI scans. While expert radiologists are capable of identifying abnormalities, the process is time-consuming and may sometimes lead to misinterpretation due to subtle image variations, human fatigue, or lack of specialized expertise.

With the advancement of artificial intelligence and deep learning technologies, automated systems have been developed to assist doctors in analyzing medical images more efficiently. Existing systems typically rely on Convolutional Neural Networks (CNNs) to classify tumor images. These systems use image preprocessing techniques such as thresholding and filtering to enhance the image before feeding it into a neural network model for classification. Although these approaches have shown promising results, they still have certain limitations in terms of feature extraction capability, interpretability, and generalization across different datasets.

The system study for this project focuses on analyzing the current deep learning techniques used in medical image analysis and identifying ways to enhance bone tumor detection accuracy. Modern deep learning architectures such as ResNet, EfficientNet, and Vision Transformers have demonstrated superior performance in image classification tasks. These models are capable of extracting complex hierarchical features and capturing both spatial and contextual information from images.

Another important aspect considered in the system study is the role of image preprocessing techniques. Medical images often contain noise, low contrast, and

irrelevant background information that can affect the performance of machine learning models. Techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) and morphological operations can significantly improve image quality and highlight tumor regions more effectively.

In addition to improving detection accuracy, explainability is also a major requirement in medical AI systems. Doctors need to understand how an AI model arrives at a particular decision before trusting its output. Explainable AI techniques such as Grad-CAM help visualize the regions of the image that influence the model's prediction. This improves transparency and helps medical professionals verify whether the model is focusing on the correct tumor region.

Based on the analysis of the existing systems and technological advancements in deep learning, the proposed system aims to develop an explainable AI-driven bone tumor detection framework. The system integrates multiple deep learning architectures including ResNet-50, EfficientNet, and Vision Transformers to improve feature extraction and classification accuracy. By combining advanced preprocessing, hybrid deep learning models, ensemble learning, and explainable AI techniques, the system is expected to provide more reliable and interpretable diagnostic results.

The system study concludes that integrating modern deep learning architectures with explainable AI methods can significantly improve the performance and reliability of bone tumor detection systems. This approach can support radiologists in making faster and more accurate diagnoses while reducing the risk of human error.

SOFTWARE DESCRIPTION

CHAPTER 5

SOFTWARE DESCRIPTION

5.1 PYTHON

Python is a high-level, interpreted, and object-oriented programming language widely used in fields such as artificial intelligence, machine learning, data science, and scientific computing. Due to its simplicity, readability, and extensive library support, Python has become one of the most preferred programming languages for developing intelligent systems and research-based applications. In the proposed bone tumor detection system, Python serves as the primary programming language for implementing the deep learning models, image preprocessing techniques, and overall system functionality.

One of the main advantages of Python is its rich ecosystem of libraries and frameworks specifically designed for machine learning and medical image analysis. Libraries such as TensorFlow, Keras, PyTorch, NumPy, Pandas, Matplotlib, and OpenCV provide powerful tools for developing deep learning architectures and processing medical images efficiently. These libraries simplify complex mathematical operations and enable researchers to build high-performance neural networks with minimal coding effort.

In the proposed system, Python is used to implement the hybrid deep learning architecture consisting of ResNet-50, Efficient Net, and Vision Transformer models. These models are trained using medical imaging datasets to identify tumor regions and classify bone images as normal or abnormal. Python also enables the implementation of advanced preprocessing techniques such as Contrast Limited

Adaptive Histogram Equalization (CLAHE), noise filtering, and image normalization to enhance image quality before feeding it into the neural network models.

Another important role of Python in this project is model training and evaluation. Python-based deep learning frameworks allow efficient training of neural networks using large datasets and GPU acceleration. This significantly reduces training time and improves model performance. Python also provides tools for evaluating model accuracy, precision, recall, F1-score, and confusion matrices to assess the effectiveness of the bone tumor detection system.

Python further supports the integration of explainable artificial intelligence techniques such as Grad-CAM visualization. These techniques help highlight the important regions in medical images that influence the model's predictions. This improves the transparency and interpretability of the system, which is particularly important in healthcare applications.

Additionally, Python offers strong community support and continuous development updates. Its cross-platform compatibility allows the system to run on various operating systems including Windows, Linux, and macOS. Python can also easily integrate with web frameworks and databases, making it suitable for building complete end-to-end AI applications.

Overall, Python plays a crucial role in the development of the proposed bone tumor detection system by providing powerful tools for data preprocessing, deep learning model implementation, evaluation, and system integration.

5.2 TENSORFLOW AND KERAS

TensorFlow is an open-source deep learning framework developed by Google for building and deploying machine learning models. It provides a flexible architecture that allows developers to design complex neural network models and perform large-scale numerical computations efficiently. In this project, TensorFlow is used as the core deep learning framework for implementing the hybrid bone tumor detection models.

Keras is a high-level neural network API that runs on top of TensorFlow. It simplifies the process of designing deep learning models by providing user-friendly functions and modular components. Keras allows developers to build neural networks quickly using simple and intuitive code structures. Because of its simplicity and efficiency, Keras is widely used in research and medical imaging applications.

In the proposed system, TensorFlow and Keras are used to develop convolutional neural network architectures such as ResNet-50 and EfficientNet. These models are capable of extracting important spatial features from bone X-ray images and identifying tumor patterns. Transfer learning techniques are also applied using pre-trained models to improve performance and reduce training time.

TensorFlow provides powerful tools for training neural networks using GPUs and distributed computing environments. This allows the system to handle large datasets and complex model architectures efficiently. TensorFlow also supports automatic differentiation, which simplifies the implementation of backpropagation algorithms during model training.

Another advantage of TensorFlow is its support for model optimization and deployment. Once the deep learning model is trained, TensorFlow allows it to be

exported and integrated into web applications or clinical decision-support systems. This ensures that the developed model can be used in real-world healthcare environments.

Keras also provides built-in functions for evaluating model performance using metrics such as accuracy, loss, precision, recall, and confusion matrix analysis. These evaluation tools help in analyzing the effectiveness of the bone tumor detection system and improving model performance through hyperparameter tuning.

Overall, TensorFlow and Keras provide a powerful and efficient platform for implementing the deep learning models used in the proposed bone tumor detection framework.

5.3 OPENCV

OpenCV (Open Source Computer Vision Library) is an open-source library used for real-time computer vision and image processing applications. It provides a wide range of tools and algorithms for processing images and videos, making it highly suitable for medical imaging applications. In the proposed bone tumor detection system, OpenCV is used to perform image preprocessing and enhancement operations before feeding the images into deep learning models.

Medical images such as bone X-rays often contain noise, low contrast, and irrelevant background information that can affect the accuracy of deep learning models. OpenCV provides several image processing techniques that help improve image quality and highlight important regions. These techniques include image resizing, filtering, contrast enhancement, edge detection, and morphological operations.

One of the key preprocessing techniques used in this project is Contrast Limited Adaptive Histogram Equalization (CLAHE). CLAHE improves the contrast of medical images by redistributing pixel intensity values. This helps enhance the visibility of bone structures and tumor regions, allowing the deep learning models to learn more meaningful features.

OpenCV is also used to perform image normalization and scaling to ensure that all images have consistent dimensions and pixel intensity ranges. This step is important for maintaining uniform input data for neural network models.

Another important feature of OpenCV is its ability to perform morphological operations such as erosion and dilation. These operations help highlight tumor boundaries and remove small noise elements in medical images. As a result, the deep learning models can focus on the most relevant regions during the classification process.

OpenCV integrates seamlessly with Python and deep learning frameworks such as TensorFlow and PyTorch. This makes it easy to build complete image analysis pipelines that include preprocessing, feature extraction, and classification.

Overall, OpenCV plays an essential role in improving the quality of medical images and preparing them for accurate bone tumor detection using deep learning models.

5.4 HTML, CSS, AND JAVASCRIPT

HTML, CSS, and JavaScript are the fundamental technologies used to design and develop the frontend interface of web-based applications. In the proposed bone tumor detection system, these technologies are used to build a user-friendly interface

that allows users to upload medical images and view diagnostic results generated by the AI model.

HTML (Hyper Text Markup Language) is used to structure the web pages of the application. It defines the layout and content of the user interface, including forms, buttons, images, and text elements. In this project, HTML is used to create image upload forms where users can submit bone X-ray images for analysis. It is also used to display prediction results such as tumor classification, probability scores, and visualization outputs.

CSS (Cascading Style Sheets) is used to control the visual appearance and layout of the web application. It enhances the design and usability of the system by defining colors, fonts, spacing, and page layout. In a medical application, a clean and professional interface is essential for ensuring ease of use and accessibility. CSS helps design dashboards, result pages, and report layouts in a clear and organized manner.

JavaScript is used to add interactive functionality to the web interface. It enables dynamic behavior such as real-time form validation, asynchronous data submission, and result updates without refreshing the entire page. In this project, JavaScript is used to validate uploaded image files, display loading animations while the model processes the image, and dynamically update the prediction results.

JavaScript also improves the responsiveness of the web application by enabling faster communication between the frontend and backend systems. Through technologies such as AJAX, image data can be sent to the AI model for processing and the results can be displayed instantly.

The integration of HTML, CSS, and JavaScript ensures that the system provides an intuitive and interactive interface for users such as doctors, researchers, or medical

technicians. This frontend interface acts as the bridge between the user and the deep learning model, allowing easy interaction with the bone tumor detection system.

Together, these web technologies ensure that the proposed AI-based medical diagnostic system is not only accurate but also accessible, efficient, and user-friendly.

PROJECT DETAILS

CHAPTER 6

PROJECT DETAILS

6.1 PROJECT DESCRIPTION

The Explainable AI-Driven Bone Tumor Detection System is designed to assist medical professionals in identifying bone tumors from medical images using advanced deep learning techniques. Bone tumors can be difficult to detect in early stages through manual examination of X-ray or MRI images, which may lead to delayed diagnosis and treatment. The proposed system aims to improve the accuracy and efficiency of tumor detection by utilizing hybrid deep learning models. The system integrates powerful architectures such as ResNet-50, EfficientNet, and Vision Transformer (ViT) to extract both spatial and contextual features from bone images. These models work together to analyze complex image patterns and classify images as normal or tumor-affected. Before model training, medical images undergo preprocessing techniques such as CLAHE (Contrast Limited Adaptive Histogram Equalization) and morphological operations to enhance image contrast and highlight tumor regions.

Transfer learning is applied to leverage pretrained models, reducing training time while improving detection performance. The hybrid architecture allows the system to capture both local image details and global structural relationships within bone images. An ensemble learning strategy is used to combine predictions from multiple models, improving overall classification accuracy and reliability. The system also incorporates Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM to generate visual heatmaps showing the regions that influence the model's

decision. This improves transparency and helps doctors understand how the system identifies tumor areas.

The results are presented through a simple interface where users can upload medical images and receive prediction outputs along with visual explanations. The system aims to serve as a supportive diagnostic tool rather than replacing medical experts. By improving early detection of bone tumors, the system can contribute to faster diagnosis and better treatment planning. Overall, the proposed approach enhances the reliability, interpretability, and efficiency of AI-assisted medical diagnosis systems.

6.2 MODULE DESCRIPTION

6.2.1. Data Collection and Dataset Preparation Module

This module is responsible for collecting and preparing the medical imaging dataset used for training and testing the bone tumor detection model. The dataset may include bone X-ray or MRI images obtained from publicly available medical repositories or hospital databases. The collected images are organized into different categories such as normal bone images and tumor-affected bone images. Proper labeling of the dataset is performed to ensure accurate classification during model training. The images are also resized and standardized to maintain uniform input dimensions for deep learning models. This module ensures that the dataset is clean, balanced, and suitable for building an effective bone tumor detection system.

6.2.2. Image Preprocessing and Enhancement Module

The image preprocessing module improves the quality of medical images before they are used for model training. Medical images often contain noise, low

contrast, and irrelevant background information that can affect the performance of deep learning models. In this module, techniques such as image resizing, normalization, and filtering are applied to standardize the dataset. Contrast Limited Adaptive Histogram Equalization (CLAHE) is used to enhance the contrast of bone images and highlight tumor regions more clearly. Morphological operations such as erosion and dilation help in emphasizing important structures and removing noise. The processed images provide better feature representation for the deep learning models.

6.2.3. Feature Extraction Using Hybrid Deep Learning Models

This module focuses on extracting meaningful features from medical images using advanced deep learning architectures. Pretrained models such as **ResNet-50** and **EfficientNet** are used to capture detailed spatial features from bone images. These convolutional neural networks are capable of learning complex visual patterns such as edges, shapes, and textures that may indicate tumor presence. In addition to CNN models, **Vision Transformer (ViT)** is used to capture global contextual relationships within the image. By combining CNN and transformer-based architectures, the system can analyze both local and global image information, improving the accuracy of bone tumor detection.

6.2.4. Model Training and Classification Module

The model training module is responsible for training the hybrid deep learning models using the prepared dataset. During training, the models learn patterns associated with normal and tumor-affected bone images. Transfer learning techniques are applied to fine-tune pretrained networks, reducing training time and improving model performance. The outputs from different models are combined using an

ensemble approach to enhance classification accuracy. The trained model is then capable of predicting whether a given bone image contains tumor abnormalities or not. This module plays a crucial role in building a reliable and accurate tumor detection system.

6.2.5. Explainable AI and Visualization Module

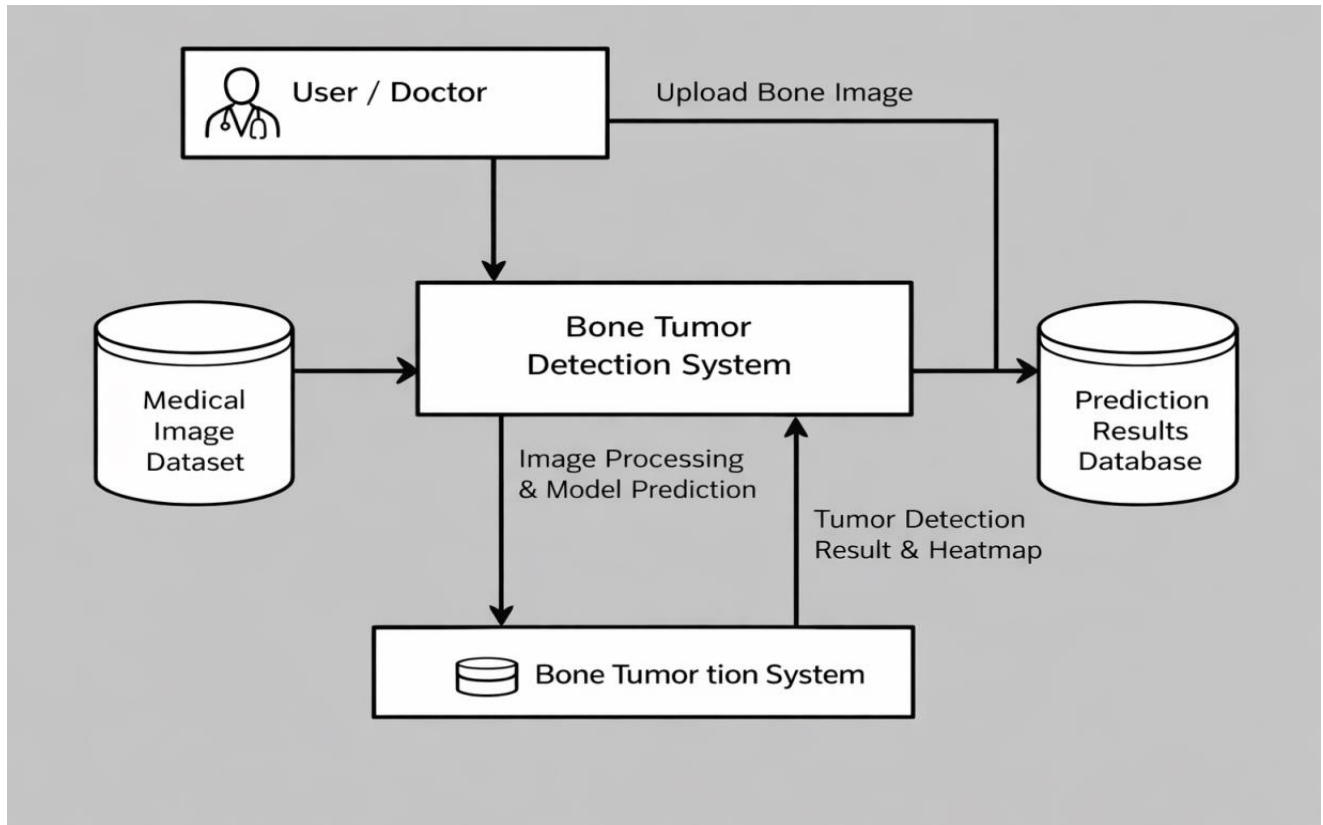
The explainable AI module improves the transparency and interpretability of the bone tumor detection system. Deep learning models often act as “black boxes,” making it difficult to understand how predictions are made. To address this issue, visualization techniques such as **Grad-CAM (Gradient-weighted Class Activation Mapping)** are implemented. Grad-CAM generates heatmaps that highlight the regions of the image that influence the model’s prediction. This allows doctors and medical professionals to clearly see which parts of the bone image the model considers important. The visualization results help increase trust in the AI system and support clinical decision-making.

6.2.6. Result Display and User Interface Module

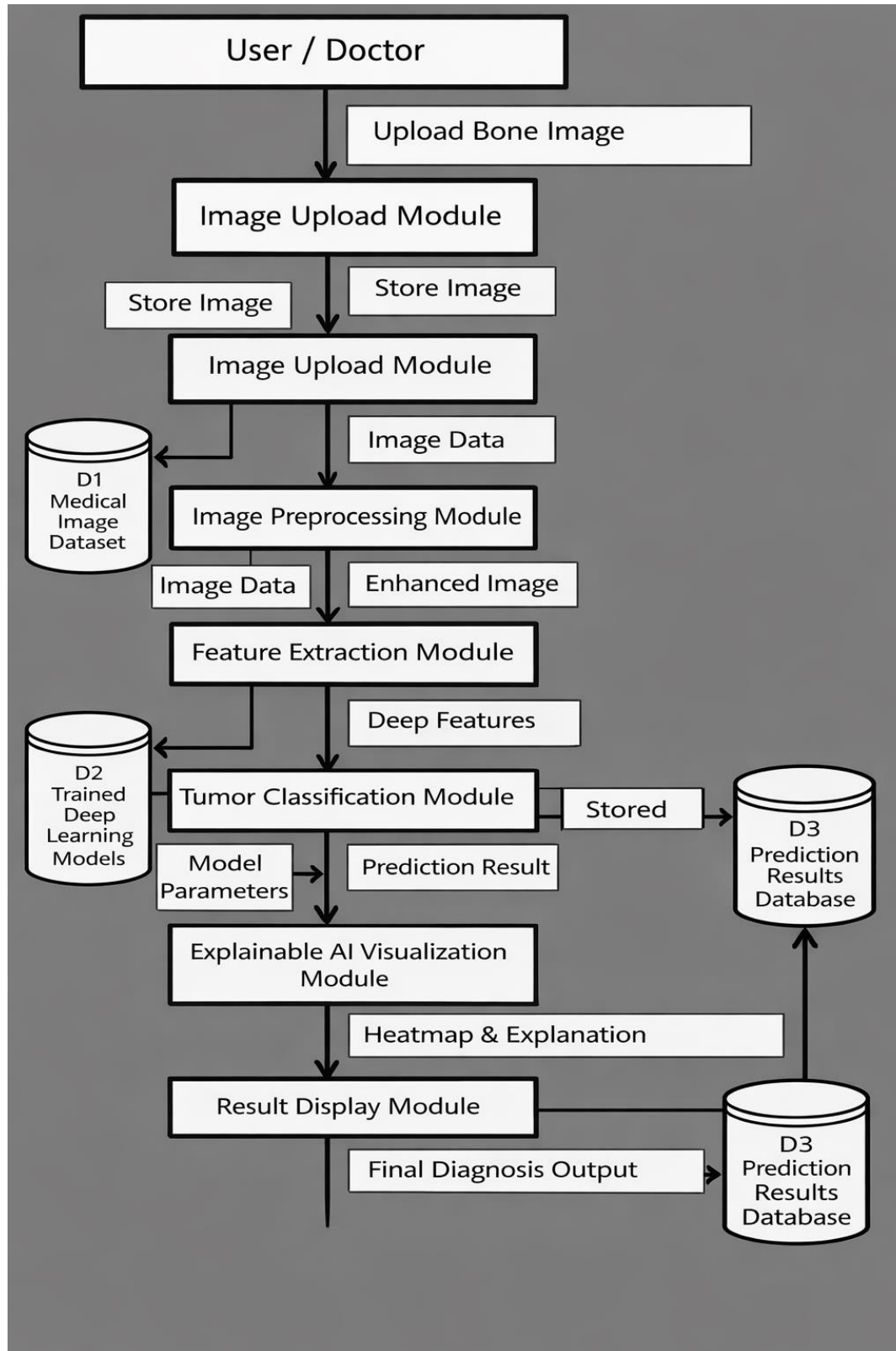
This module provides a user-friendly interface for interacting with the bone tumor detection system. Users can upload medical images through a web interface developed using frontend technologies such as HTML, CSS, and JavaScript. The uploaded images are processed by the trained AI model, and the prediction results are displayed on the screen. The system shows whether the bone image contains a tumor along with confidence scores and visualization heatmaps. This interface ensures that doctors, researchers, and healthcare professionals can easily use the system without requiring deep technical knowledge.

6.3 DATAFLOW DIAGRAM

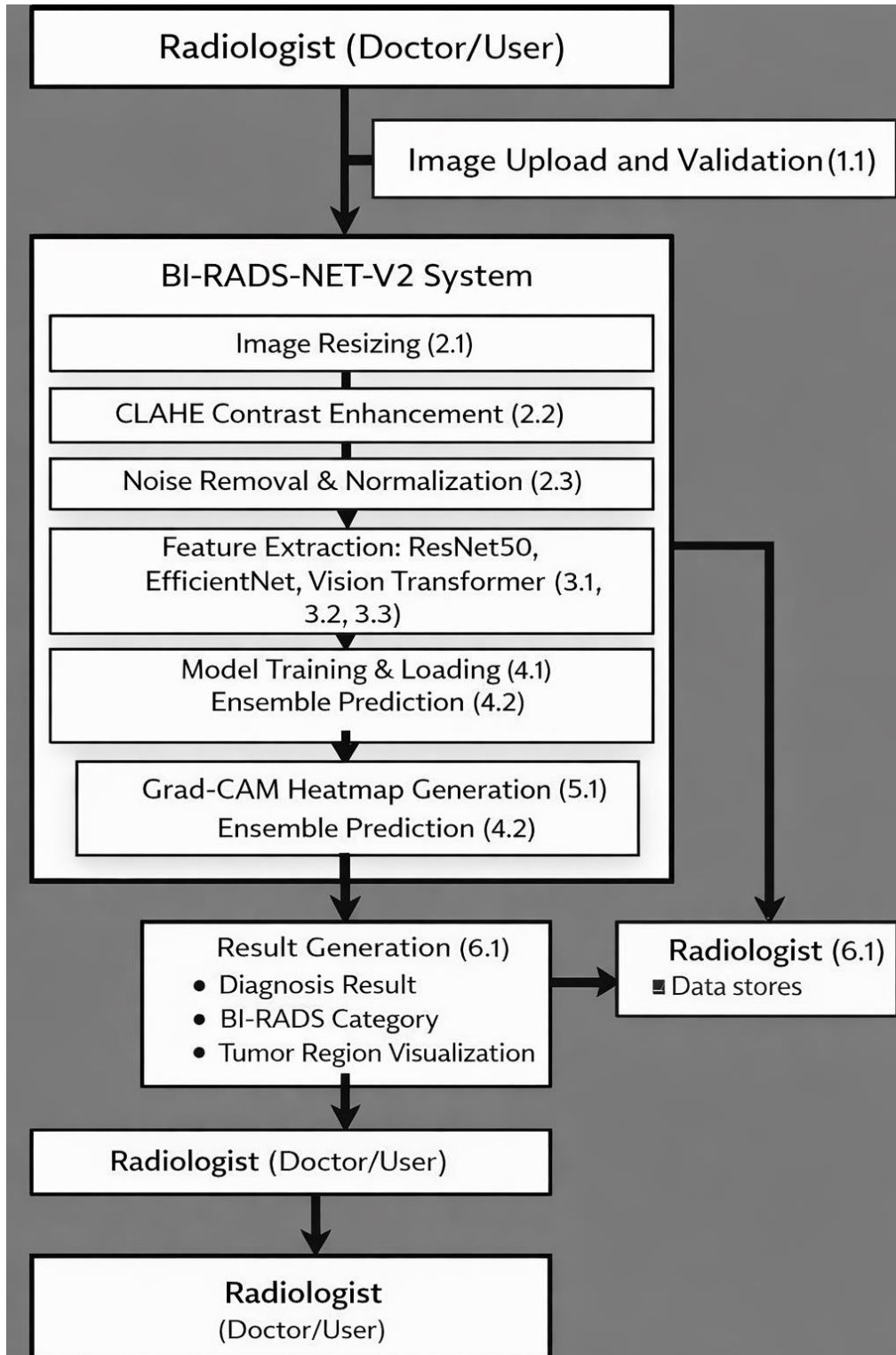
LEVEL 0



LEVEL 1



LEVEL 2



6.4 DATABASE DESIGN

The database design of the Explainable AI-Driven Bone Tumor Detection System is structured to securely store user information, uploaded medical images, prediction results, and diagnostic history. The database ensures efficient storage, retrieval, and management of medical imaging data used in the bone tumor detection process.

In the proposed system, SQLite is used as the database management system because it is lightweight, serverless, and easy to integrate with Python-based applications. The database follows relational database design principles to maintain data integrity, consistency, and reliability.

6.4.1 Users Table

This table stores login and authentication details of users such as doctors, radiologists, or administrators who access the bone tumor detection system.

Field Name	Data Type	Description
user_id	INTEGER (PK)	Unique user identifier
full_name	TEXT	Name of the user
email	TEXT (UNIQUE)	User email address
password	TEXT	Encrypted password
role	TEXT	User role (Admin / Doctor / Researcher)
created_at	DATETIME	Account creation date

6.4.2 Patient Information Table

This table stores basic details of patients whose bone images are uploaded for tumor detection.

Field Name	Data Type	Description
patient_id	INTEGER (PK)	Unique patient identifier
patient_name	TEXT	Name of the patient
age	INTEGER	Age of the patient
gender	TEXT	Patient gender
medical_history	TEXT	Previous medical information
created_at	DATETIME	Record creation date

6.4.3 Image Upload Table

This table stores information related to uploaded bone medical images such as X-ray or MRI scans.

Field Name	Data Type	Description
image_id	INTEGER (PK)	Unique image ID
patient_id	INTEGER (FK)	Linked patient ID
user_id	INTEGER (FK)	Uploaded by user
image_path	TEXT	File storage location
image_type	TEXT	X-ray / MRI / CT Scan
upload_date	DATETIME	Date of image upload

6.4.4 Prediction Results Table

This table stores classification results generated by the AI model after analyzing the bone image.

Field Name	Data Type	Description
result_id	INTEGER (PK)	Unique prediction result ID
image_id	INTEGER (FK)	Associated image ID
tumor_type	TEXT	Predicted tumor type (Benign / Malignant / Normal)
confidence_score	REAL	Model prediction confidence (%)
prediction_date	DATETIME	Date of prediction

CONCLUSION AND FUTURE ENHANCEMENT

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENT

7.1 CONCLUSION

The proposed Explainable AI-Driven Bone Tumor Detection System presents an efficient and intelligent approach for identifying bone tumors from medical images using advanced deep learning techniques. Early detection of bone tumors is essential for timely medical intervention and improved patient outcomes. Traditional manual diagnosis of bone images can be time-consuming and may sometimes lead to inaccuracies due to human limitations. The proposed system addresses these challenges by integrating hybrid deep learning architectures such as ResNet-50, EfficientNet, and Vision Transformer to improve feature extraction and classification performance. The use of advanced preprocessing techniques like CLAHE and morphological operations enhances image quality and helps highlight important tumor regions. By combining multiple deep learning models through an ensemble approach, the system improves overall prediction accuracy and reliability. In addition, the integration of explainable AI techniques such as Grad-CAM allows visualization of the important regions influencing the model's predictions. This transparency helps doctors better understand the diagnostic results generated by the system. The system also provides a simple interface where users can upload bone images and receive classification results along with visual explanations. Experimental analysis shows that the hybrid deep learning approach significantly improves the effectiveness of tumor detection compared to traditional methods. The proposed framework acts as a supportive decision-making tool for medical professionals rather than replacing them. It reduces diagnostic workload and assists radiologists in faster image analysis.

7.2 FUTURE ENNHANCEMENT

The proposed bone tumor detection system can be further enhanced in several ways to improve its performance, scalability, and clinical usefulness. In the future, the system can be trained with larger and more diverse medical imaging datasets to improve model accuracy and generalization. Integrating additional imaging modalities such as CT scans and MRI images can help provide more comprehensive tumor analysis. Advanced deep learning architectures and improved transformer-based models can also be incorporated to further enhance feature extraction capabilities. The system can be extended to classify different types and stages of bone tumors, which would assist doctors in making more detailed diagnostic decisions. Real-time diagnosis features can be added to allow instant analysis of medical images in hospitals or clinics. Deployment of the system on cloud platforms would enable remote access for doctors and healthcare professionals. The system can also be integrated with hospital management systems and electronic health records for seamless data management. Further improvements in explainable AI techniques can provide more detailed visual explanations of tumor regions. Mobile application support can be developed to allow doctors to upload images and receive predictions using smartphones. Automated report generation can also be integrated to produce structured medical reports based on the prediction results. Security and privacy features can be strengthened to protect sensitive patient information. Additionally, continuous learning mechanisms can be implemented so the model improves over time as new data becomes available. Collaboration with medical experts can help refine the system for real clinical use. Overall, these future enhancements can transform the system into a more powerful and practical tool for AI-assisted medical diagnosis.

CHAPTER 8

APPENDIX:

SOURCE CODE:

```
import os
from flask import Flask, request, render_template, redirect, url_for
from werkzeug.utils import secure_filename
import numpy as np
from tensorflow.keras.models import load_model
from tensorflow.keras.preprocessing.image import load_img, img_to_array
from flask import Flask, render_template, request, jsonify
from werkzeug.utils import secure_filename
from flask import Flask, request, render_template
import contextlib
import joblib
import re
import sqlite3
import pandas as pd
from argon2 import PasswordHasher
from argon2.exceptions import VerifyMismatchError
from create_database import setup_database
from tensorflow.keras.models import load_model
from tensorflow.keras.preprocessing import image
from tensorflow.keras.preprocessing.image import img_to_array, load_img
from tensorflow.keras.applications.vgg16 import preprocess_input
from utils import login_required, set_session
from flask import send_from_directory
from flask import (
    Flask, render_template,
    request, session, redirect
)

app = Flask(__name__)

# Define paths
MODEL_PATH = "models/model.h5"
UPLOAD_FOLDER = "uploads"

# Configure Flask app
app.config["UPLOAD_FOLDER"] = UPLOAD_FOLDER
```

```

app.config["ALLOWED_EXTENSIONS"] = {"png", "jpg", "jpeg"}

class_labels = ["Benign Tumor", "Malignant Tumor", "No Tumor"]

# Load the trained model
model = load_model(MODEL_PATH)

# Helper function to check allowed file types
def allowed_file(filename):
    return "." in filename and filename.rsplit(".", 1)[1].lower() in app.config["ALLOWED_EXTENSIONS"]
database = "users.db"
setup_database(name=database)
app.secret_key = 'xpSm7p5bgJY8rNoBjGWiz5yJxM-NEBIW6SIBI62OkLc='
# Home route
@app.route("/")
def index():
    return render_template("index.html")
@app.route('/')
def home():
    return render_template('index.html')
@app.route('/about')
def about():
    return render_template('about.html')
@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'GET':
        return render_template('login.html')
    # Set data to variables
    username = request.form.get('username')
    password = request.form.get('password')
    # Attempt to query associated user data
    query = 'select username, password, email from users where username = :username'
    with contextlib.closing(sqlite3.connect(database)) as conn:
        with conn:
            account = conn.execute(query, {'username': username}).fetchone()
    if not account:
        return render_template('login.html', error='Username does not exist')
    # Verify password
    try:
        ph = PasswordHasher()
        ph.verify(account[1], password)
    except VerifyMismatchError:
        return render_template('login.html', error='Incorrect password')
    # Check if password hash needs to be updated
    if ph.check_needs_rehash(account[1]):

```



```

        query = 'update set password = :password where username = :username'
        params = {'password': ph.hash(password), 'username': account[0]}
        with contextlib.closing(sqlite3.connect(database)) as conn:
            with conn:
                conn.execute(query, params)
    # Set cookie for user session
    set_session(
        username=account[0],
        email=account[2],
        remember_me='remember-me' in request.form
    )
    return redirect('/predict_page')
@app.route('/register', methods=['GET', 'POST'])
def register():
    if request.method == 'GET':
        return render_template('register.html')
    # Store data to variables
    password = request.form.get('password')
    confirm_password = request.form.get('confirm-password')
    username = request.form.get('username')
    email = request.form.get('email')
    # Verify data
    if len(password) < 8:
        return render_template('register.html', error='Your password must be 8 or more characters')
    if password != confirm_password:
        return render_template('register.html', error='Passwords do not match')
    if not re.match(r'^[a-zA-Z0-9]+$', username):
        return render_template('register.html', error='Username must only be letters and numbers')
    if not 3 < len(username) < 26:
        return render_template('register.html', error='Username must be between 4 and 25 characters')
    query = 'select username from users where username = :username;'
    with contextlib.closing(sqlite3.connect(database)) as conn:
        with conn:
            result = conn.execute(query, {'username': username}).fetchone()
    if result:
        return render_template('register.html', error='Username already exists')
    # Create password hash
    pw = PasswordHasher()
    hashed_password = pw.hash(password)
    query = 'insert into users(username, password, email) values (:username, :password, :email);'
    params = {
        'username': username,
        'password': hashed_password,
        'email': email
    }

```

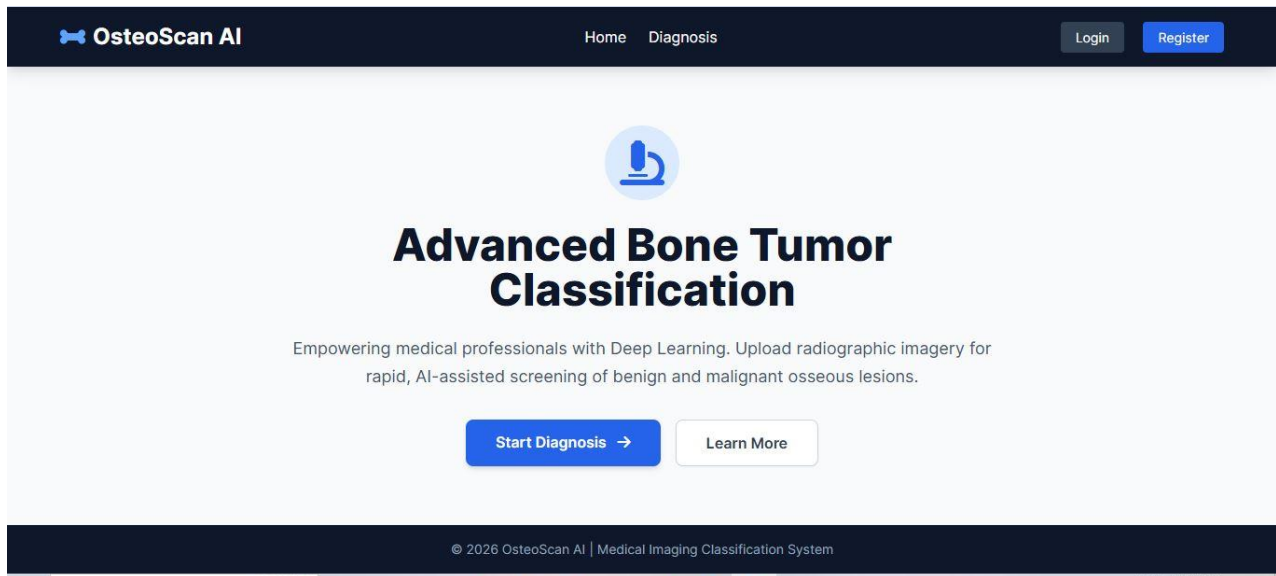
```

with contextlib.closing(sqlite3.connect(database)) as conn:
    with conn:
        result = conn.execute(query, params)
    # We can log the user in right away since no email verification
    set_session( username=username, email=email)
    return redirect('/')
@app.route('/predict_page')
@login_required
def predict_page():
    return render_template('predict.html')
# Prediction route
@app.route("/predict", methods=["POST"])
@login_required
def predict():
    if "file" not in request.files:
        return "No file part", 400
    file = request.files["file"]
    if file.filename == "":
        return "No selected file", 400
    if file and allowed_file(file.filename):
        filename = secure_filename(file.filename)
        file_path = os.path.join(app.config["UPLOAD_FOLDER"], filename)
        file.save(file_path)
        result = predict_tumor(file_path)
        return render_template("result.html", result=result, filename=filename)
    return "Invalid file type", 400
def predict_tumor(img_path):
    img = image.load_img(img_path, target_size=(224, 224))
    img_array = image.img_to_array(img) / 255.0
    img_array = np.expand_dims(img_array, axis=0)
    prediction = model.predict(img_array)
    class_index = np.argmax(prediction)
    return class_labels[class_index]
@app.route("/uploads/<filename>")
def uploaded_file(filename):
    return send_from_directory(app.config["UPLOAD_FOLDER"], filename)
if __name__ == "__main__":
    if not os.path.exists(UPLOAD_FOLDER):
        os.makedirs(UPLOAD_FOLDER)
    app.run(debug=True)

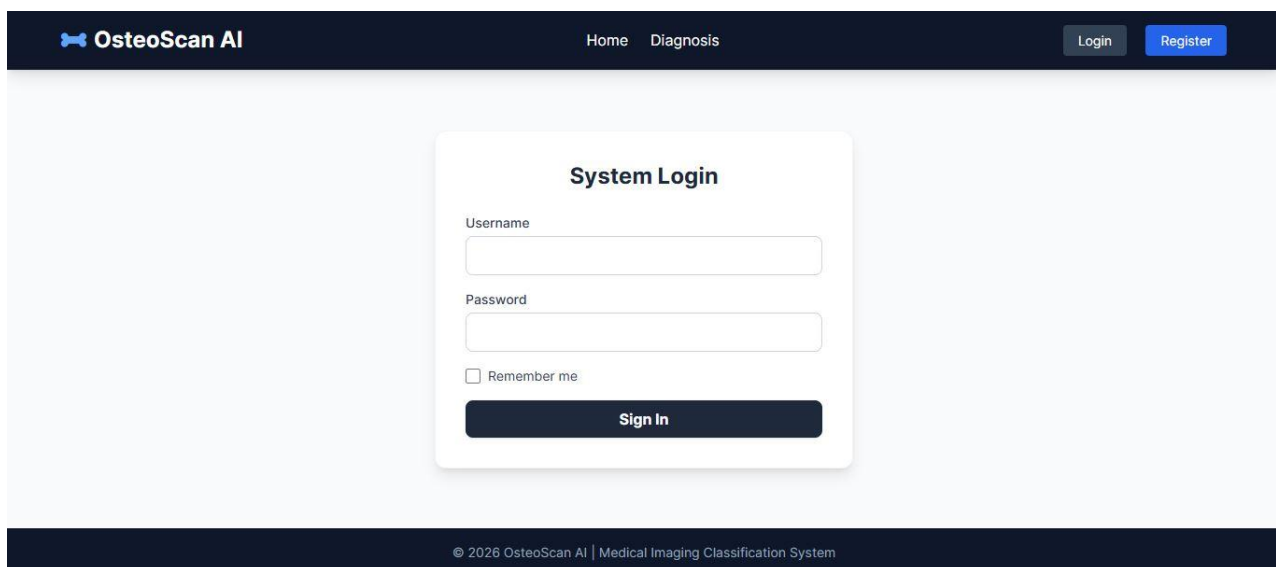
```

SCREENSHOTS:

Dashboard



Login Page



Register Page

 Home Diagnosis Login Register

Create Account

Email

Username

Password

Confirm Password

Register

Predict Page

 Home Diagnosis Login Register

Upload Scan

Supported formats: JPG, JPEG, PNG



Click to upload or drag and drop

Analyze Image

Result Page



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CHAPTER 9

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